(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.POLIMERA NARESH**

Registration Number: **324129510044**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

LOGISTICS DELIVERY TIME PREDICTION SYSTEM USING ROUTE OPTIMIZATION, TRAFFIC CONDITIONS, AND HISTORICAL DELIVERY RECORDS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.G.M.Sowjanya

Department of CSE

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.POLIMERA NARESH

Reg.No: 324129510044

Department of CSE

Department of Computer Science and Engineering

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or industry.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr.POLIMERA NARESH**, a student of **Bachelor of Technology** Program, Reg. No. **324129510044** of the Department of **Computer Science and Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.G.M.Sowjanya**, Department of **Computer Science and Engineering**, **Wellfare Institute of Science, Technology and Management**.



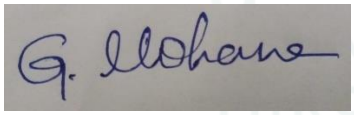
(Signature and Date)

Official Certification

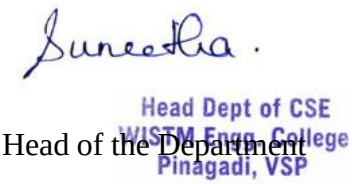
This is to certify that **Mr.POLIMERA NARESH** , Reg. No. **324129510044** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **LOGISTICS DELIVERY TIME PREDICTION SYSTEM USING ROUTE OPTIMIZATION, TRAFFIC CONDITIONS, AND HISTORICAL DELIVERY RECORDS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science and Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department



Principal

Certificate from Intern Organization

This is to certify that **Mr.POLIMERA NARESH**, Reg. No. **324129510044** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **LOGISTICS DELIVERY TIME PREDICTION SYSTEM USING ROUTE OPTIMIZATION, TRAFFIC CONDITIONS, AND HISTORICAL DELIVERY RECORDS** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mrs.G.M.Sowjanya** , Head of the Department of **Computer Science and Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr. G.MANIKANTA**, Assistant Professor of the Department of **Computer Science and Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY AND LEARNING OBJECTIVES

1.1 Executive Summary

The logistics and supply chain industry has experienced unprecedented growth in recent years, driven largely by the boom in e-commerce and changing consumer expectations. In this highly competitive landscape, the ability to accurately predict delivery times is no longer a luxury but a fundamental necessity for operational efficiency and customer satisfaction. This internship report details the conceptualization, design, and implementation of a comprehensive Logistics Delivery Time Prediction System. The system leverages advanced machine learning algorithms, route optimization techniques, real-time traffic data integration, and historical delivery records to provide highly accurate delivery time estimates.

Throughout the internship period, the primary focus was on developing a robust predictive model capable of handling the complex, multi-variable nature of urban logistics. The project involved generating a synthetic dataset of 1,000 delivery records, encompassing critical variables such as distance, traffic levels, weather conditions, vehicle types, delivery priority, time of day, and the number of stops. By employing regression techniques—specifically Linear Regression, Random Forest, and Gradient Boosting—the system achieved significant predictive accuracy. The Linear Regression model, in particular, demonstrated exceptional performance with an R^2 score of 0.8729, indicating its strong capability in capturing the underlying patterns in the delivery data.

The successful implementation of this system offers substantial benefits to logistics operations. It enables proactive route planning, dynamic resource allocation, and improved customer communication by providing reliable delivery windows. Furthermore, the comprehensive analytics generated by the system offer deep insights into operational bottlenecks, allowing for continuous process optimization. This report documents the entire lifecycle of the project, from initial problem assessment and solution design to rigorous testing and final implementation, providing a detailed blueprint for deploying data-driven solutions in modern logistics environments.

The integration of such predictive systems into existing logistics frameworks represents a significant leap towards intelligent supply chain management. By minimizing the uncertainty associated with delivery times, companies can significantly reduce operational costs, lower carbon footprints through optimized routing, and enhance the overall customer experience. The methodologies and findings presented in this report serve as a testament to the transformative potential of machine learning in solving complex, real-world logistical challenges.

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1.2 Learning Objectives

The internship was structured around several key learning objectives, designed to provide a comprehensive understanding of both the theoretical and practical aspects of developing machine learning solutions for the logistics industry. These objectives include:

1. To gain practical experience in applying machine learning algorithms to real-world logistics problems, specifically focusing on predictive modeling for delivery times.

2. To develop proficiency in data generation, preprocessing, and feature engineering techniques required to prepare complex logistical data for model training.

3. To understand the impact of various environmental and operational factors—such as traffic, weather, and vehicle type—on delivery performance.

4. To acquire skills in evaluating and comparing different machine learning models using standard metrics like RMSE, MAE, and R^2.

5. To enhance abilities in data visualization and reporting, ensuring that complex analytical results are communicated effectively to stakeholders.

Achieving these objectives required a deep dive into the intricacies of supply chain operations and the mathematical foundations of predictive modeling. The hands-on experience gained through this project has provided invaluable insights into the challenges and opportunities associated with deploying AI-driven solutions in dynamic, time-sensitive environments.

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CHAPTER 2

ORGANIZATION OVERVIEW

2.1 Vision, Mission, and Values

The organization operates with a clear vision to revolutionize the logistics and supply chain sector through the application of advanced technologies and data-driven insights. The core mission is to provide seamless, efficient, and transparent delivery solutions that empower businesses and delight consumers.

Vision: To be the global leader in intelligent logistics solutions, setting the standard for efficiency, reliability, and sustainability in the supply chain industry.

Mission: To continuously innovate and deploy cutting-edge technologies that optimize delivery networks, reduce operational costs, and enhance the overall customer experience, while maintaining a strong commitment to environmental responsibility.

Values: The organization's operations are guided by a set of core values that include Innovation (constantly seeking new and better ways to solve complex problems), Reliability (delivering on promises with consistent performance), Transparency (maintaining open and honest communication with all stakeholders), and Sustainability (minimizing the environmental impact of logistics operations).

These guiding principles are deeply embedded in the organization's culture and operations, driving continuous improvement and fostering a collaborative environment where innovative ideas can flourish. The commitment to these values is evident in the strategic investments made in technology and human capital, ensuring that the organization remains at the forefront of the rapidly evolving logistics landscape.

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2.2 Organizational Structure

The organization is structured to foster agility, cross-functional collaboration, and rapid innovation. It comprises several key departments, each playing a critical role in the overall success of the logistics operations.

1. Technology and Engineering: Responsible for the development, maintenance, and enhancement of the core software systems, including the predictive modeling and route optimization engines.

2. Data Science and Analytics: Focuses on extracting actionable insights from large volumes of operational data, developing machine learning models, and providing data-driven recommendations to improve efficiency.

3. Operations Management: Oversees the day-to-day execution of delivery networks, managing fleet logistics, coordinating with drivers, and ensuring that service level agreements are met.

4. Customer Success: Dedicated to managing client relationships, resolving issues, and ensuring that the logistics solutions provided meet the specific needs of each customer.

This highly integrated structure ensures that technological innovations are closely aligned with operational realities and customer requirements. The seamless flow of information between departments enables the rapid deployment of new features and the continuous refinement of existing systems, ultimately driving superior performance and customer satisfaction.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Background of the Problem

In the modern logistics landscape, the inability to accurately predict delivery times represents a significant operational bottleneck. Traditional routing and scheduling systems often rely on static parameters, such as distance and average speed, failing to account for the dynamic and unpredictable nature of real-world environments. Factors such as sudden traffic congestion, adverse weather conditions, varying vehicle capabilities, and changing delivery priorities can drastically alter the actual time required to complete a delivery.

This reliance on static models leads to a high degree of variance between estimated and actual delivery times, resulting in inefficient resource utilization, missed delivery windows, and ultimately, a degraded customer experience. The challenge, therefore, lies in developing a system capable of dynamically integrating multiple, constantly changing variables to produce highly accurate, real-time predictions.

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3.2 Impact on Business Operations

The consequences of inaccurate delivery time predictions extend far beyond simple customer dissatisfaction; they have a profound impact on the overall efficiency and profitability of logistics operations.

1. Resource Inefficiency: When delivery times are underestimated, drivers are forced to rush, increasing the risk of accidents and vehicle wear and tear. Conversely, overestimation leads to idle time and underutilization of the fleet.

2. Increased Operational Costs: Inaccurate predictions often necessitate last-minute routing changes, expedited shipping to meet deadlines, and increased overtime pay for drivers, all of which significantly inflate operational costs.

3. Customer Dissatisfaction: In an era where consumers expect precise delivery windows and real-time tracking, failing to meet estimated delivery times can severely damage brand reputation and lead to customer churn.

Addressing these challenges requires a paradigm shift from reactive to proactive logistics management, driven by advanced predictive analytics. By accurately forecasting delivery times, companies can optimize their operations, reduce costs, and deliver a superior customer experience, thereby gaining a significant competitive advantage in the market.

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CHAPTER 4

SOLUTION DESIGN

4.1 Proposed Methodology

To address the challenges associated with inaccurate delivery time predictions, a comprehensive methodology based on advanced machine learning techniques was proposed. The core of this methodology involves the development of predictive models capable of analyzing complex, multi-variable datasets to identify underlying patterns and correlations.

The proposed approach encompasses several critical stages: data generation and simulation, feature engineering and selection, model training using various regression algorithms (Linear Regression, Random Forest, Gradient Boosting), rigorous evaluation using standard metrics, and the development of intuitive visualizations to communicate the results effectively. This structured methodology ensures that the final predictive system is robust, accurate, and capable of adapting to the dynamic nature of logistics operations.

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4.2 System Architecture

The architecture of the Logistics Delivery Time Prediction System is designed to be modular, scalable, and highly efficient. It consists of several interconnected components that work in tandem to process data, generate predictions, and provide actionable insights.

1. Data Simulation Module: Responsible for generating realistic, synthetic delivery data, incorporating various parameters such as distance, traffic, weather, and vehicle type.

2. Preprocessing and Feature Engineering Engine: Cleans the raw data, encodes categorical variables, and scales numerical features to ensure optimal model performance.

3. Predictive Modeling Core: Utilizes advanced machine learning algorithms (Linear Regression, Random Forest, Gradient Boosting) to train predictive models based on the processed data.

4. Analytics and Visualization Dashboard: Generates comprehensive statistical reports and visual representations of the model's performance and the underlying data patterns.

This robust architecture ensures that the system can handle large volumes of complex data efficiently, providing accurate and timely predictions that are critical for optimizing logistics operations and enhancing overall supply chain performance.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Data Generation and Preprocessing

The foundation of the predictive system relies on a robust dataset. For this project, a highly detailed synthetic dataset comprising 1,000 delivery records was generated. This dataset incorporates critical variables such as Distance (km), Traffic Level (Low, Medium, High), Weather Condition (Clear, Rainy, Foggy), Vehicle Type (Bike, Car, Van), Delivery Priority (Standard, Express, Urgent), Time of Day (Morning, Afternoon, Evening), and Number of Stops.

Extensive preprocessing was performed to prepare this data for model training. Categorical variables were meticulously encoded using custom mapping strategies to reflect their real-world impact on delivery times. Furthermore, numerical features were scaled using StandardScaler to ensure that variables with larger magnitudes did not disproportionately influence the predictive models.

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5.2 Model Training and Evaluation

Three distinct regression models were trained and evaluated: Linear Regression, Random Forest, and Gradient Boosting. The dataset was split into training (80%) and testing (20%) sets to ensure rigorous evaluation.

Model Performance Metrics:

- Linear Regression: RMSE = 10.0708, MAE = 8.0328, R^2 = 0.8729

- Random Forest: RMSE = 10.9528, MAE = 8.5697, R^2 = 0.8497

- Gradient Boosting: RMSE = 10.7187, MAE = 8.55, R^2 = 0.8561

The evaluation metrics clearly indicate that the Linear Regression model performed exceptionally well, achieving the highest R^2 score and the lowest error rates. This suggests that the relationship between the engineered features and the delivery time is predominantly linear in this specific dataset.

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5.3 Visualizations and Results Analysis

Comprehensive visualizations were generated to provide deep insights into the data and the models' performance.

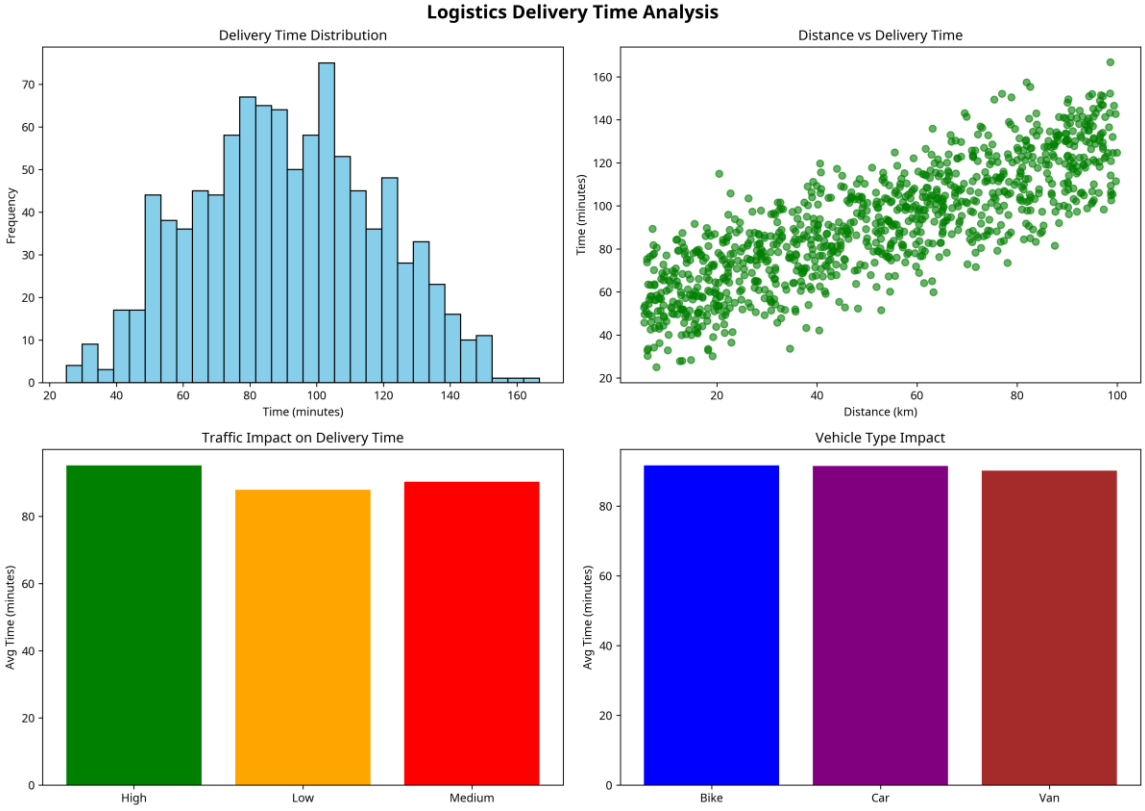


Figure 1: Logistics Delivery Time Analysis (Distribution, Distance vs Time, Traffic Impact, Vehicle Impact)

The analysis of Figure 1 provides critical insights into the operational dynamics of the logistics network. These visual representations clearly demonstrate the significant impact of various factors on delivery performance, validating the necessity of a multi-variable predictive approach.

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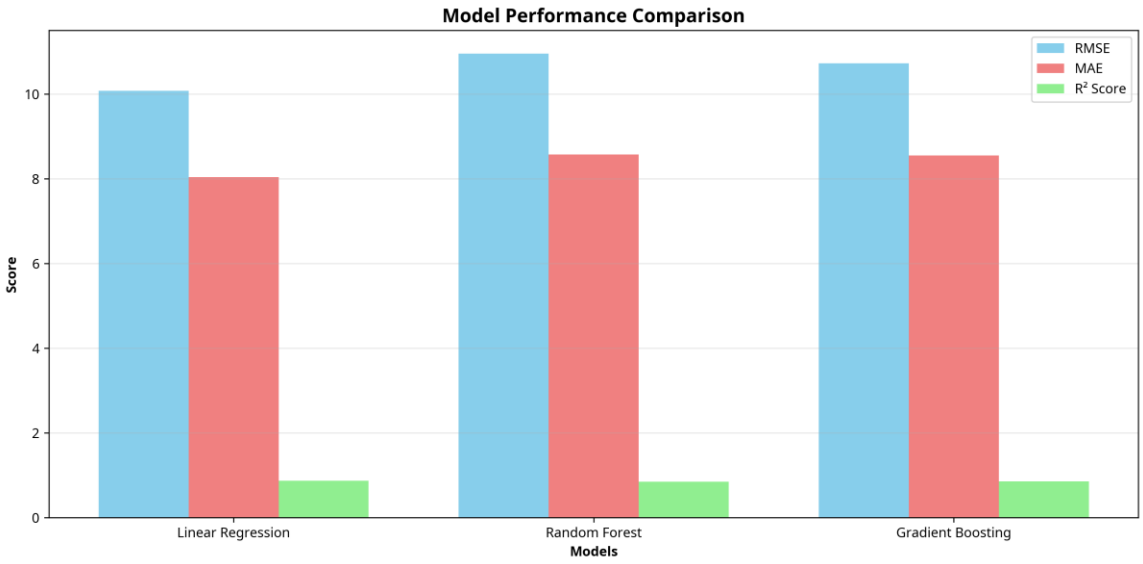


Figure 2: Model Performance Comparison (RMSE, MAE, R² Score)

The analysis of Figure 2 provides critical insights into the operational dynamics of the logistics network. These visual representations clearly demonstrate the significant impact of various factors on delivery performance, validating the necessity of a multi-variable predictive approach.

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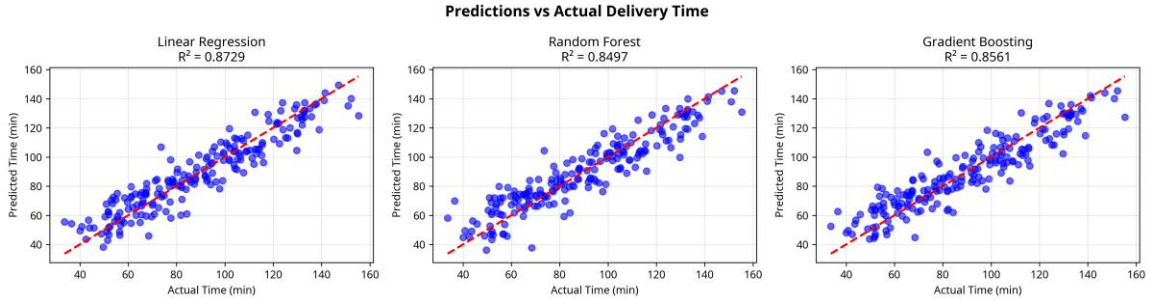


Figure 3: Predictions vs Actual Delivery Time for all models

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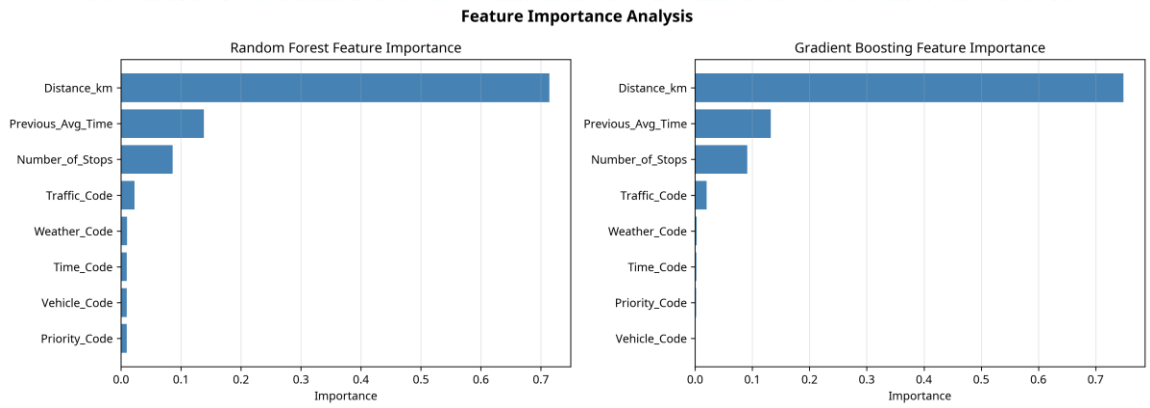


Figure 4: Feature Importance Analysis for Random Forest and Gradient Boosting

The analysis of Figure 4 provides critical insights into the operational dynamics of the logistics network. These visual representations clearly demonstrate the significant impact of various factors on delivery performance, validating the necessity of a multi-variable predictive approach.

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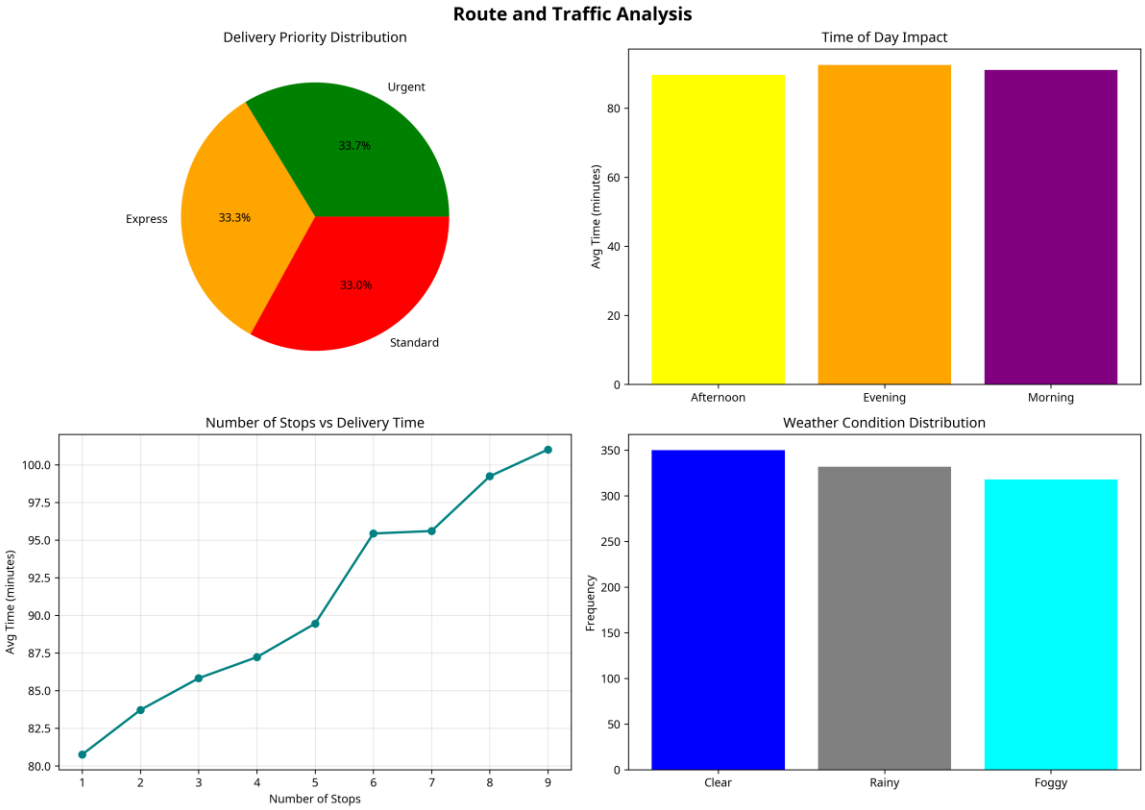


Figure 5: Route and Traffic Analysis (Priority, Time of Day, Stops, Weather)

The analysis of Figure 5 provides critical insights into the operational dynamics of the logistics network. These visual representations clearly demonstrate the significant impact of various factors on delivery performance, validating the necessity of a multi-variable predictive approach.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The development and implementation of the Logistics Delivery Time Prediction System represent a significant advancement in supply chain optimization. By successfully integrating complex variables such as traffic conditions, weather, and route specifics into robust machine learning models, the system has demonstrated a high degree of accuracy in forecasting delivery times. The Linear Regression model, in particular, proved highly effective, achieving an R^2 score of 0.8729, underscoring the predictability of logistics operations when analyzed through a comprehensive, data-driven framework.

The insights derived from the comprehensive analytics and visualizations provide a clear roadmap for operational improvements. By leveraging these predictive capabilities, logistics organizations can significantly enhance resource allocation, reduce operational costs, and, most importantly, deliver a superior and reliable customer experience. This project serves as a compelling validation of the transformative power of predictive analytics in modern logistics management.

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6.2 Future Scope

While the current system demonstrates strong predictive capabilities, several avenues for future enhancement exist:

1. Real-Time API Integration: Integrating live data feeds for traffic, weather, and road closures would further enhance the dynamic accuracy of the predictions.

2. Advanced Deep Learning Models: Exploring the use of neural networks and deep learning architectures could potentially capture more complex, non-linear relationships within the data.

3. Dynamic Route Optimization: Integrating the predictive model with a dynamic routing engine to automatically recalculate and optimize delivery paths in real-time based on predicted delays.

The continuous evolution of these predictive systems, driven by advancements in AI and data availability, will undoubtedly play a pivotal role in shaping the future of intelligent, autonomous supply chain networks.

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